

Continuum cone spectra: a symmetric-cone classification and a nonpolyhedral obstruction

Candidate partial solution for arXiv:1107.4532

11 August 2026

Status. Candidate theorem, likely valid; a complete answer within finite-dimensional symmetric cones and a strict obstruction to the naive general answer. A separate question in the source was answered negatively in the literature by Gripenberg in 2015.

1 Questions and outcomes

Source. B. Lemmens and R. Nussbaum, *Continuity of the cone spectral radius*, arXiv:1107.4532 [1]. For a homogeneous order-preserving map $f : C \rightarrow C$, its cone spectrum is

$$\sigma_C(f) = \{\lambda \geq 0 : f(x) = \lambda x \text{ for some } 0 \neq x \in C\}.$$

The source asks (i) whether the Bonsall radius $\tilde{r}_C(f)$ always equals the supremal orbital growth radius $r_C(f)$ for general nonlinear maps, and (ii) which finite-dimensional closed cones admit a map whose cone spectrum contains a continuum.

Literature answer to (i). Gripenberg's Example 3.1 is a continuous, order-preserving, degree-one homogeneous map on a closed pointed convex cone in a Banach space. His Lemma 3.5 proves

$$\tilde{r}_C(f) \geq 1 \quad \text{and} \quad r_C(f) \leq \frac{1}{2},$$

so the answer is negative [2]. This directly resolves the open question on p. 2 of the source.

For (ii), the source proves that every polyhedral cone has finite cone spectrum for every order-preserving homogeneous map, and constructs an infinite spectrum on every nonpolyhedral cone. It does not decide when the spectrum can contain a nondegenerate compact connected set (equivalently, an interval, since the spectrum lies in $\mathbb{R}$). We prove a complete answer for symmetric cones and show that nonpolyhedrality alone is insufficient.

2 The symmetric-cone classification

Let V be a finite-dimensional Euclidean Jordan algebra with trace inner product $\langle \cdot, \cdot \rangle$, unit e , and cone of squares $\mathcal{K} = \{x^2 : x \in V\}$. This is a symmetric cone. Powers of elements of $\mathcal{K}$ are defined by spectral calculus; see [3, 4].

Lemma 1 (Löwner–Heinz at exponent $1/2$). *If $0 \leq x \leq y$ in $\mathcal{K}$, then $x^{1/2} \leq y^{1/2}$.*

Proof. First suppose x, y are interior. Inversion reverses the symmetric-cone order. Indeed, with P the quadratic representation, $x \leq y$ gives $e \leq z = P(x^{-1/2})y$. Spectral calculus gives $z^{-1} \leq e$; applying the positive map $P(x^{-1/2})$ and using $(P(x^{-1/2})y)^{-1} = P(x^{1/2})y^{-1}$ yields $y^{-1} \leq x^{-1}$.

For scalars $s > 0$,

$$\sqrt{s} = \frac{1}{\pi} \int_0^\infty t^{-1/2} \frac{s}{s+t} dt.$$

The same identity holds by Jordan spectral calculus. Since

$$x \circ (x + te)^{-1} = e - t(x + te)^{-1},$$

inverse order reversal shows that the integrand is order-preserving in x . Integration proves the claim in the interior, and replacing x, y by $x + \varepsilon e, y + \varepsilon e$ and letting $\varepsilon \downarrow 0$ proves the boundary case. $\square$

Fix a in the interior of $\mathcal{K}$, put $\ell_a(x) = \langle a, x \rangle$, and define

$$F_a(x) = (\ell_a(x)x)^{1/2} \quad (x \in \mathcal{K}). \quad (1)$$

Lemma 2 (Exact spectrum of the square-root map). *F_a is continuous, order-preserving, and homogeneous of degree one, and*

$$\sigma_{\mathcal{K}}(F_a) = \{\sqrt{\langle a, c \rangle} : c \neq 0 \text{ is a Jordan idempotent}\}. \quad (2)$$

Proof. Continuity is spectral calculus, and $F_a(tx) = tF_a(x)$ for $t \geq 0$. The cone is self-dual, so $x \leq y$ implies $\ell_a(x) \leq \ell_a(y)$. Moreover,

$$\ell_a(y)y - \ell_a(x)x = \ell_a(y)(y - x) + (\ell_a(y) - \ell_a(x))x \in \mathcal{K}.$$

Lemma 1 therefore proves order preservation.

Write $x = \sum_i \mu_i c_i$, with $\mu_i \geq 0$ and (c_i) a Jordan frame. If $F_a(x) = \lambda x$ and $x \neq 0$, then $\lambda > 0$ because a is interior. Squaring gives

$$\ell_a(x)\mu_i = \lambda^2 \mu_i^2 \quad \text{for every } i.$$

All positive μ_i are consequently equal, so $x = qc$ for $q > 0$ and a nonzero idempotent c . Direct substitution gives $F_a(qc) = \sqrt{\langle a, c \rangle} qc$. This proves both inclusions in (2). $\square$

Theorem 3 (Complete answer for symmetric cones). *For a finite-dimensional symmetric cone $\mathcal{K}$, the following are equivalent.*

1. $\mathcal{K}$ is nonpolyhedral.
2. There is a continuous order-preserving homogeneous map $F : \mathcal{K} \rightarrow \mathcal{K}$ whose cone spectrum contains a nondegenerate interval.

When (1) holds, F may be chosen in the explicit form (1).

Proof. If $\mathcal{K}$ is polyhedral, Theorem 5.5 of the source says that every such spectrum is finite, so (2) is impossible.

Conversely, decompose V into simple Jordan factors. If every factor has rank one, $\mathcal{K}$ is an orthant and hence polyhedral. Thus a nonpolyhedral $\mathcal{K}$ has a simple factor of rank at least two. Choose a Jordan frame $c_1, c_2, \dots$ in that factor. The Peirce space V_{12} is nonzero; choose $z \in V_{12}$ normalized by $z^2 = c_1 + c_2$. A direct use of the Peirce multiplication rules shows that

$$c(\theta) = \cos^2 \theta c_1 + \sin^2 \theta c_2 + \sin \theta \cos \theta z, \quad 0 \leq \theta \leq \frac{\pi}{2},$$

satisfies $c(\theta)^2 = c(\theta)$ and has trace one, so it is primitive.

Choose a interior whose coefficients at c_1, c_2 are $0 < \alpha < \beta$ and whose other Jordan-frame coefficients are positive. Peirce orthogonality gives

$$\langle a, c(\theta) \rangle = \alpha \cos^2 \theta + \beta \sin^2 \theta.$$

Lemma 2 now shows that $[\sqrt{\alpha}, \sqrt{\beta}] \subseteq \sigma_{\mathcal{K}}(F_a)$. $\square$

This includes Lorentz cones, real/complex/quaternionic positive-semidefinite matrix cones, and the exceptional symmetric cone. It generalizes the 2×2 real positive-semidefinite example in the source from one cone to the entire symmetric-cone class.

3 Nonpolyhedrality is not sufficient in general

The following obstruction explains why Theorem 3 does not settle every finite-dimensional cone.

Proposition 4. *There is a closed pointed full-dimensional nonpolyhedral cone $C \subset \mathbb{R}^3$ such that $\sigma_C(f)$ is countable for every order-preserving homogeneous map $f : C \rightarrow C$, even without assuming continuity.*

Proof. In the unit circle of $\mathbb{R}^2$, let

$$S = \{(\cos(1/n), \sin(1/n)) : n \geq 1\} \cup \{(1, 0), (-1, 0), (0, -1)\}, \quad B = \text{conv}(S).$$

The set S is compact. Every point of S is exposed in B : the tangent line to the unit disk at that point meets the disk only there. Conversely, Milman's converse to Krein–Milman gives $\text{ext}(B) \subseteq S$. Thus B has countably infinitely many

extreme points. A proper face of a planar compact convex set is either an extreme singleton or a line-segment edge determined by two extreme endpoints, so B has only countably many faces.

Take the cone over this base,

$$C = \{t(u, 1) : t \geq 0, u \in B\} \subset \mathbb{R}^3.$$

It is closed, pointed, and full-dimensional. It is nonpolyhedral because it has infinitely many extreme rays, while its faces—and hence its parts—are countable. Lemma 4.3 and Corollary 5.4 of the source show that two eigenvectors in the same part must have the same eigenvalue. Each part therefore contributes at most one point to $\sigma_C(f)$, proving countability. $\square$

Thus “nonpolyhedral” cannot be the answer to the source’s general problem. The unresolved classification must distinguish the much finer face/extension geometry of arbitrary cones.

4 Verification, novelty, and scope

The order-preserving calculation for F_a is explicit. Its exact spectrum follows coefficient-by-coefficient from Jordan spectral decomposition. The Peirce path is checked by $c_i \circ z = z/2$, $c_1 \circ c_2 = 0$, and $z^2 = c_1 + c_2$. Proposition 4 uses only elementary planar convexity plus the source’s same-part eigenvalue lemma.

A bounded search through 11 August 2026 found Gripenberg’s literature answer, the source’s matrix example, its later book presentation, and papers on the different complementarity notion of cone spectrum. No source found states Theorem 3, the exact formula (2), or Proposition 4. This is a bounded novelty check, not an exhaustive claim.

Scope. Theorem 3 is a full classification only for symmetric cones. Proposition 4 proves that the general finite-dimensional question requires more than polyhedral versus nonpolyhedral. No full classification of arbitrary cones is claimed.

References

- [1] B. Lemmens and R. Nussbaum, *Continuity of the cone spectral radius*, [arXiv:1107.4532](#); Proc. Amer. Math. Soc. 141 (2013), 2741–2754.
- [2] G. Gripenberg, *On the definition of the cone spectral radius*, Proc. Amer. Math. Soc. 143 (2015), 1617–1625, [doi:10.1090/S0002-9939-2014-12375-6](#).
- [3] M. S. Gowda, *Some majorization inequalities induced by Schur products in Euclidean Jordan algebras*, [arXiv:1911.00579](#).
- [4] J. Faraut and A. Korányi, *Analysis on Symmetric Cones*, Oxford University Press, 1994.